

# Building a Virtual Private Cloud on AWS Cloud

A virtual private cloud (VPC) has become increasingly popular, driven by its ability to provide secure, flexible, and scalable networking environments in the cloud. Here's a step-by-step guide to setting up a VPC service and deploying a web server using the Amazon Web Services (AWS) platform.

A virtual private cloud (VPC) is a secure, isolated segment of a public cloud where users can provision and manage resources within a virtual network environment. It offers a logically separated network space on a public cloud platform (such as AWS, Azure, or Google Cloud), enabling users to launch, control, and manage

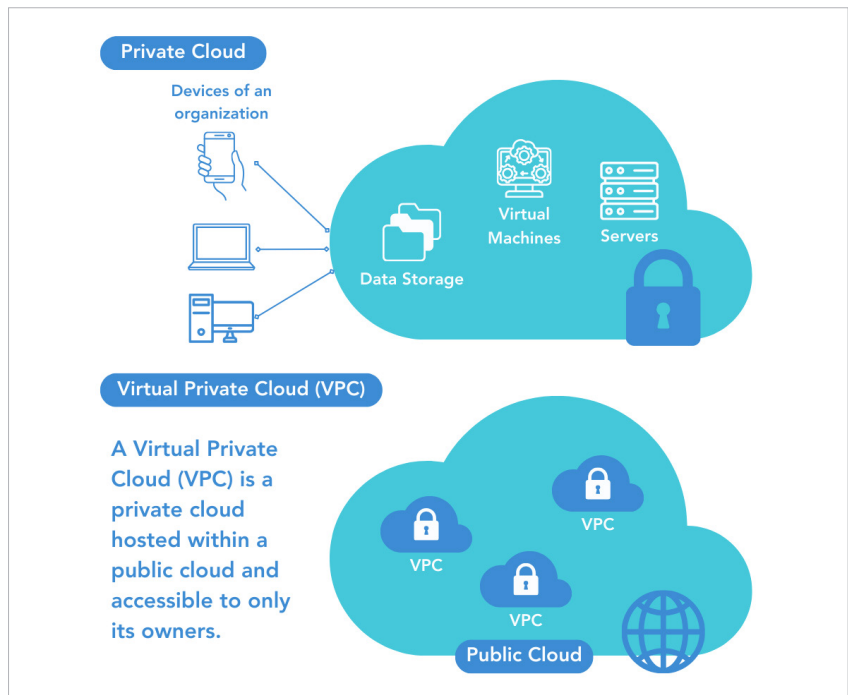


Figure 1: Introduction to virtual private cloud

Table 1: Differences between a VPC and a private cloud

| Criteria                 | Virtual private cloud (VPN)                                                                                                                                                                | Private cloud                                                                                                                                                                   |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition               | A dedicated public cloud segment where resources are allocated within a virtual network environment.                                                                                       | A dedicated cloud environment owned or operated by a single organisation.                                                                                                       |
| Infrastructure ownership | The infrastructure is owned and managed by a public cloud provider (e.g., AWS, Azure, GCP).                                                                                                | The infrastructure is owned and managed by the organisation itself.                                                                                                             |
| Resource allocation      | Resources are shared across multiple tenants but isolated within a dedicated virtual network.                                                                                              | Resources are exclusively dedicated to a single organisation, providing full environmental control.                                                                             |
| Cost structure           | A pay-as-you-go strategy reduces upfront capital costs.                                                                                                                                    | Needs a significant initial investment and continuous expenditure on infrastructure and hardware maintenance.                                                                   |
| Security and compliance  | It provides a high level of security with network isolation, encryption, and access controls, but within a shared public cloud.                                                            | It offers the highest level of security, as all resources are dedicated and controlled by a single organisation. It is suitable for sensitive data and compliance requirements. |
| Use cases                | Suitable for businesses looking for cost-effective, scalable solutions with some level of security and control. Commonly used for web applications, testing, and development environments. | Ideal for organisations with strict regulatory compliance and security requirements, such as healthcare, finance, and government sectors.                                       |